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Lab Sheet 1—Solutions to Tasks 1 and 2

2025 Formative Exercises

1 One-time encryption

Consider a one-time encryption scheme called Enc_1 where messages are bit-strings, and each bit is sampled according to the following distribution $\Pr[m = 0] = 1/4$ and $\Pr[m = 1] = 3/4$. Keys are also bit-strings, sampled freshly for each message, and of equal length to a message, with the following distribution $\Pr[k = 0] = 1/2$ and $\Pr[k = 1] = 1/2$.

Consider another one-time encryption scheme called Enc_2 where messages are bit-strings, and each bit is sampled according to the following distribution $\Pr[m = 0] = 1/2$ and $\Pr[m = 1] = 1/2$. Keys are also bit-strings, sampled freshly for each message, and of equal length to a message, with the following distribution $\Pr[k = 0] = 1/4$ and $\Pr[k = 1] = 3/4$.

Consider yet another one-time encryption scheme called Enc_3 where messages are bit-strings, and each bit is sampled according to the following distribution $\Pr[m = 0] = 1/3$ and $\Pr[m = 1] = 2/3$. Keys are also bit-strings, sampled freshly for each message, and of equal length to a message, with the following distribution $\Pr[k = 0] = 1/4$ and $\Pr[k = 1] = 3/4$.

a. For each of the schemes sample a 12 bit message from the respective message space adhering to the given distribution. Then sample a corresponding key string adhering to the given distribution, and encrypt the message using the key. What do you notice about the distribution of the bits in the ciphertext for each scheme?

b. Work out, using the given distributions, what is $\Pr[M = m|C = c]$.

c. What would adherence to Kerckhoff's principle (regarding the no security by obscurity) imply in relation to the three constructions?

d. Are any of the encryption schemes secure and why? Can you prove perfect secrecy for any/some/all of them according to Shannon's definition?

Solution

There are a few facts that are useful in the context of this example. These are:

$$\Pr[M = m|C = c] = \frac{\Pr[M = m \wedge C = c]}{\Pr[C = c]} \tag{1}$$

and

$$\Pr[M = m | C = c] = \frac{\Pr[C = c | M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \tag{2}$$

and, if (and only if) two random variables are independent we have that $\Pr[X = x \wedge Y = y] = \Pr[X = x] \cdot \Pr[Y = y]$. Because we pick secret keys independently from plaintexts we can therefore conclude that

$$\Pr[M = m \wedge K = k] = \Pr[M = m] \cdot \Pr[K = k]. \tag{3}$$

In the following I provide a full solution for Enc_1 and a full solution for Enc_2. You should be able to use these to get to your own solution for Enc_3.

Scheme Enc_1

I start with some general observations: the scheme is a one to one match to the one-time pad encryption scheme. By this I not only mean that the encryption function is the same, but that it respects the important choice of uniformly random keys. Thus the expectation is that we will find it to be secure according to Shannon's definition.

a. When you are asked to sample from a distribution, then this means that you draw elements from its support according to some rules that imply how likely you pick certain elements. Thus a nice sample from a distribution here looks like: $m = 000011111111$, and $k = 010101010101$, and this means $c = 010110101010$. There are as many zero bits as there are one bits in c.

b. You are asked to produce certain conditional probabilities, and you might recognise that this specific conditional probability is part of the security notion for perfect secrecy. How do you calculate these given the information that you have? There are multiple options: doing it directly via (1) or doing the conditional probability the "other way around" using Bayes (2). The first option requires to produce the joint distribution $\Pr[M = m \wedge C = c]$. This requires a bit of work because M and C cannot be assumed to be independent: it is precisely that what the proof for the one-time pad shows based on the properties of encryption and other assumptions. This is done via first working out the other conditional probability, which is perhaps slightly more "natural" because $\Pr[C = c | M = m]$ simply means "what is the probability to see c if m was the message". We also know that $\Pr[C = c | M = m] = \Pr[C = c \wedge M = m] \cdot \Pr[M = m]$ (we are given $\Pr[M = m]$, and we can get to the unknown joint probabilities using the encryption rule:

	$m = 0$	$m = 1$
$k = 0$	$c = 0$	$c = 1$
$k = 1$	$c = 1$	$c = 0$

This table is made according to the one-time pad encryption rule, but it is simplified to a single bit. This is a useful trick because in a scheme where l bits get encrypted, the same rule is applied to each bit independently, and thus reasoning about a single bit is sufficient. Now we need to add the provided information into that table (i.e. all the distribution information plus the rules that give c given m and k:

	$\Pr[m = 0] = 1/4$	$\Pr[m = 1] = 3/4$
$\Pr[k = 0] = 1/2$	$\Pr[m = 0 \wedge k = 0]$	$\Pr[m = 1 \wedge k = 0]$
$\Pr[k = 1] = 1/2$	$\Pr[m = 0 \wedge k = 1]$	$\Pr[m = 1 \wedge k = 1]$

We know the probabilities for the key and message, and now we need to work out the probabilities for the ciphertext. Recall that the key is picked independently from the message and therefore (3) holds, which means we can produce the probabilities in the table easily:

	$\Pr[m = 0] = 1/4$	$\Pr[m = 1] = 3/4$
$\Pr[k = 0] = 1/2$	$\Pr[m = 0 \wedge k = 0] = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$	$\Pr[m = 1 \wedge k = 0] = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$
$\Pr[k = 1] = 1/2$	$\Pr[m = 0 \wedge k = 1] = \frac{1}{8}$	$\Pr[m = 1 \wedge k = 1] = \frac{3}{8}$

This table represents the joint distribution between M and K. You will notice that summing across the rows gives $\Pr[K = k]$ and summing across the columns gives $\Pr[M = m]$. For instance $\Pr[M = 0] = \sum_k \Pr[M = 0, K = k] = \Pr[M = 0, K = 0] + \Pr[M = 0, K = 1] = 1/8 + 1/8 = 1/4$.

We know turn to the conditional distribution $\Pr[K = k|M = m]$. We know that $c = m + k$. Therefore

$$
\begin{aligned}
\Pr[C = c|M = m] =& \frac{\Pr[c = m + K \wedge M = m]}{\Pr[M = m]} = \\
& \frac{\Pr[c = m + K] \cdot \Pr[M = m]}{\Pr[M = m]} = \\
& \Pr[c = m + K].
\end{aligned}
$$

The first step holds because of the encryption rule, and the rule (1), the second step holds because we have fixed c, m and therefore the only random variable is K which is independent of M. The resulting probability $\Pr[c = m + K]$ is now a probability over the choice of K (we fixed m and c). Therefore for all combinations m, k we select the probability of $K = k$ with $c = m + k$. Because there is only one value of k possible for each pair m, c, and all choices for k are equally likely we find that

$$
\begin{aligned}
\Pr[C = 0|M = 0] &= \Pr[K = 0] = 1/2 \\
\Pr[C = 0|M = 1] &= \Pr[K = 1] = 1/2 \\
\Pr[C = 1|M = 0] &= \Pr[K = 1] = 1/2 \\
\Pr[C = 1|M = 1] &= \Pr[K = 0] = 1/2
\end{aligned}
$$

From these we can also work out the marginal probability $\Pr[C = c] = \sum_M \Pr[C = c|M = m] \cdot \Pr[M = m]$, which gives $Pr[C = 0] = 1/2$ and $\Pr[C = 1] = 1/2$.

The task is to produce $\Pr[M = m|C = c]$ so we now use Bayes theorem (2) to flip the conditionals, and we use that $\Pr[C|M] = \Pr[C]$, and find:

$$\begin{aligned}
\Pr[M = 0|C = 0] &= \tfrac{\Pr[C=0|M=0]\cdot\Pr[M=0]}{\Pr[C=0]} = \Pr[M = 0] = 1/4 \\
\Pr[M = 0|C = 1] &= \tfrac{\Pr[C=1|M=0]\cdot\Pr[M=0]}{\Pr[C=1]} = \Pr[M = 0] = 1/4 \\
\Pr[M = 1|C = 0] &= \tfrac{\Pr[C=0|M=1]\cdot\Pr[M=1]}{\Pr[C=0]} = \Pr[M = 1] = 3/4 \\
\Pr[M = 1|C = 1] &= \tfrac{\Pr[C=1|M=1]\cdot\Pr[M=1]}{\Pr[C=1]} = \Pr[M = 1] = 3/4.
\end{aligned}$$

c. Adhering to Kerckhoff's principle (no security by obscurity means) that an adversary knows all parameters of the encryption scheme, only the choice of k is unknown. Therefore they know that 0 bits are more likely to occur in the ciphertext than 1 bits.

d. This scheme satisfies the requirement that $\Pr[M|C] = \Pr[M]$ (or equivalently that $\Pr[C|M] = \Pr[M]$). We showed that this is true for all choices of $M = M$ and $C = c$ so we provided a full proof.

Scheme Enc_2

The immediate general observation here is that the keys are not chosen from a uniform distribution. This implies immediately that the scheme cannot be secure according to Shannon's definition.

a. A nice sample from a distribution here looks like: $k = 000011111111$, and $m = 010101010101$, and this means $c = 010110101010$. There are as many zero bits as there are one bits in c.

b. Without repeating the reasoning from before, here are the relevant tables. The setup for this scheme is nearly identical to what we did for Enc_1 except that we have swapped the distributions for key and message spaces.

	$\Pr[m = 0] = 1/2$	$\Pr[m = 1] = 1/2$
$\Pr[k = 0] = 1/4$	$\Pr[m = 0 \wedge k = 0]$	$\Pr[m = 1 \wedge k = 0]$
$\Pr[k = 1] = 3/4$	$\Pr[m = 0 \wedge k = 1]$	$\Pr[m = 1 \wedge k = 1]$

Filling in the probabilities we get:

	$\Pr[m = 0] = 1/2$	$\Pr[m = 1] = 1/2$
$\Pr[k = 0] = 1/4$	$\Pr[m = 0 \wedge k = 0] = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$	$\Pr[m = 1 \wedge k = 0] = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$
$\Pr[k = 1] = 3/4$	$\Pr[m = 0 \wedge k = 1] = \frac{3}{8}$	$\Pr[m = 1 \wedge k = 1] = \frac{3}{8}$

Also in this example, it holds that $\Pr[K = k|M = m] = \Pr[c = m + K]$, which means that the distribution of $C|M$ is due to the distribution of K.

$$\begin{aligned}
\Pr[C = 0|M = 0] &= \Pr[K = 0] = 1/4 \\
\Pr[C = 0|M = 1] &= \Pr[K = 1] = 3/4 \\
\Pr[C = 1|M = 0] &= \Pr[K = 1] = 3/4 \\
\Pr[C = 1|M = 1] &= \Pr[K = 0] = 1/4
\end{aligned}$$

We notice that $\Pr[C = 0] = \Pr[C = 0|M = 0] \cdot \Pr[M = 0] + \Pr[C = 0|M = 1] \cdot \Pr[M = 1] = 1/2$. This is squarely because our messages are from a uniform distribution! If this was not the case, which is exactly what happens in Enc_3, then the ciphertext bits would not look like random garbage.

At this point we have proof that the scheme is not secure because we have just found that $\Pr[C|M] \neq \Pr[C]$ (which is an equivalent to Shannon's criteria that $\Pr[M|C] = \Pr[M]$, which you can convince yourself of by using Bayes theorem).

Next we calculate the conditionals that are asked for by noticing that $\Pr[C] = \Pr[M]$:

$$\begin{aligned}
\Pr[M = 0|C = 0] &= \tfrac{\Pr[C=0|M=0] \cdot \Pr[M=0]}{\Pr[C=0]} = 1/4 \\
\Pr[M = 0|C = 1] &= \tfrac{\Pr[C=1|M=0] \cdot \Pr[M=0]}{\Pr[C=1]} = 3/4 \\
\Pr[M = 1|C = 0] &= \tfrac{\Pr[C=0|M=1] \cdot \Pr[M=1]}{\Pr[C=0]} = 3/4 \\
\Pr[M = 1|C = 1] &= \tfrac{\Pr[C=1|M=1] \cdot \Pr[M=1]}{\Pr[C=1]} = 1/4.
\end{aligned}$$

We find again that $\Pr[M|C] \neq Pr[M]$ which is proof that the scheme is not secure. This hopefully shows you that even though the ciphertext distribution is uniform, the scheme still leaks information.

Let's make this visible by repeatedly encrypting the same message using Enc_2. Suppose we encrypt 0010 multiple times with different one-time keys:

k	$m = 0010$
1110	$c = 1100$
1011	$c = 1001$
0111	$c = 0101$
1101	$c = 1111$

You can see that each ciphertext looks like random garbage. But in fact, each ciphertext gives a lot of information about the message. This is because if you see a 1 bit in the ciphertext, you know with high probability that the message bit was 0. This is because there is a high chance that the corresponding key bit was 1 (because of the key distribution), and therefore the message bit must have been 0. As an adversary, if we know that the same message was encrypted multiple times, we can see from the ciphertexts, that the first bit is nearly always 1 (three out of four times), which is exactly the key distribution, thus with near certainty the key bit here was 1 and therefore the message bit was 0. You can observe that we can apply the same kind of reasoning for each ciphertext bit individually, and therefore, we get with near certainty the message in this case.

c. Adhering to Kerchhoffs' principle means that an adversary has high chance to decrypt despite the one-time key use.

d. We have already answered this question via the conditionals: the scheme is not secure.

2 The operation $x \oplus c$ is a bijection

We used a key concept to argue about the security of the one-time pad, and in fact any construction that relies on a Boolean addition. This key concept is that the Boolean addition $x \oplus c$, with $x, c \in \{0,1\}^l$ and c being a constant, is a bijection for all choices of c.

We also used a property of bijections, which is that if they go from a pre-image space $\mathcal{X}$, with a uniform probability distribution $p_{\mathcal{X}}$, to an image space $\mathcal{Y}$, then the distribution $p_{\mathcal{Y}}$ must also be uniform.

What is a bijection?. Why does a bijection preserve the uniform distribution distribution, i.e. if the elements in $\mathcal{X}$ occur with a uniform distribution, then the elements in $\mathcal{Y}$ also have a uniform distribution? Consider the simplest case where $\mathcal{X} = \{0,1\}$, and c is thus also just a single bit. In what sense does a bijection preserve a distribution in general?

Solution

A function $f(x)$ is a bijection if and only if it is invertible, i.e. there is a function $g()$ such that $g(f(x)) = x$ and also $f(g(y)) = y$.

The function $f(x) = x \oplus c$ (with c being some constant value) is invertible because there is the function $g(y) = y \oplus c$ which satisfies the property from before: $y = x \oplus c$ and so $g(y) = (x \oplus c) \oplus c = x$ (this was $g(f(x)) = x$). Equally $f(g(y)) = f(y \oplus c) = (y \oplus c) \oplus c = y$.

Another way of visualising what is a bijection is to consider it as a map between the pre-image space and the image space. This map is "one-to-one": each element x in the pre-image space is mapped to exactly one element y in the image space.

In the case of a Boolean function and with a constant c, the c determines how the map from pre-image to image space (and reverse) is defined. In particular, if $c = 0$, then 0 is mapped to 0 and 1 is mapped to 1. If $c = 1$, the 0 is mapped to 1 and 1 is mapped to 0. Thus depending on c we either map an element onto itself or onto the other element.

Now let's look at what happens when we consider that the distribution on the pre-image space is such that $\Pr[X = 0] = p$ and thus $\Pr[X = 1] = 1 - p$.

	$\Pr[X = 0] = p$	$\Pr[X = 1] = 1 - p$
$c = 0$	$\Pr[Y = 0 \oplus 0] = p$	$\Pr[Y = 1 \oplus 0] = 1 - p$
$c = 1$	$\Pr[Y = 1] = p$	$\Pr[Y = 0] = 1 - p$

If $c = 0$ then the bijection leads to the probabilities $\Pr[X = 0] = \Pr[Y = 0] = p$ and $\Pr[X = 1] = \Pr[Y = 1] = 1 - p$. Therefore, irrespective of the value of p we assign the same probabilities to the events $X = x$ and $Y = y$.

If $c = 1$ then we have that $\Pr[X = 0] = \Pr[Y = 1] = p$ and $\Pr[X = 1] = \Pr[Y = 0] = 1 - p$. This bijection flips the assignment of the probabilities. However, we retain the characteristic of the distribution, which is that there is one element that occurs with probability p and the other element occurs with probability $1 - p$. So in this sense the bijection has mapped the distribution on X to Y.

Only for the specific choice of $p = 1/2$ it holds that $\Pr[X = 0] = \Pr[X = 1] = \Pr[Y = 0] = \Pr[Y = 1] = 1/2$ for both $c = 0$ and $c = 1$.

Therefore only for $p = 1/2$, which is the uniform distribution, we can say that the distribution on X is the same as the distribution on Y, whereby "the same" now really means that $\Pr[X = 0] = \Pr[Y = 0]$ and $\Pr[X = 1] = \Pr[Y = 1]$.